

About Us

Helios Dynamics is a technology startup working on next-generation hazard detection systems. We develop practical detection and intelligence systems for the Armed Forces — solutions that combine sensors, signal processing, and robust communications to reveal hazards that other tools miss. We're already collaborating with the Army and Paramilitary Forces on this program and several parallel projects, rapidly advancing prototypes through lab work and operational trials.

Role Overview:

We are looking for an **Intern** (preferably **EXTC / Electronics**) to join our R&D team. The intern will play a key role in researching and developing sensor technologies and running experiments to identify the most effective configurations for accurate hazard detection.

Key Responsibilities:

- Research and evaluate sensor technologies for:
 - **Metal detection** (e.g., GPR, EMI, magnetometers)
 - **Electric field sensing** (electrostatic/electromagnetic sensors)
 - **Chemical/explosive trace detection** (e.g., electronic noses, spectroscopy, IMS)
- Compare sensors based on feasibility, accuracy, cost, and environmental performance.
- Design and conduct **testing frameworks** to explore permutations and combinations of sensors to maximize detection accuracy.
- Assist in prototyping and integration of sensors into hardware platforms.
- Document findings, prepare reports, and present research outcomes to the technical team.

Skills & Qualifications:

- Currently pursuing or recently completed a degree in **Electronics, EXTC, Instrumentation, or related fields**.
- Strong grasp of **circuits, sensors, and signal processing fundamentals**.
- Familiarity with **RF systems, electromagnetics, and/or chemical sensing methods** is a plus.
- Proficiency in programming for data analysis and signal processing (**Python, MATLAB, or C/C++**).
- Experience with hardware platforms (Arduino, Raspberry Pi, microcontrollers) is a must.
- Strong analytical mindset with problem-solving skills.
- Ability to work independently and communicate findings clearly.

Internship Duration & Compensation:

- 45 Days: To be extended upon further consideration
- Compensation: Between Rs. 3,500 – Rs. 5,000 per month
- Mode of Internship: Remote; with weekly meetings for updates

What We Offer:

- Internship completion Certificate from the company.
- Hands-on exposure to advanced sensor research and working problem statements given by defense organizations.
- Guidance and mentorship from experienced professionals.